CORE PURE / FURTHER CALCULUS

Improper Integrals and Mean Values

Questions by topic 16 questions · 116 marks

Questions

Question 1	*4 marks*	2
Question 2	*5 marks*	3
Question 3	*8 marks*	4
Question 4	*6 marks*	5
Question 5	*6 marks*	6
Question 6	*8 marks*	7
Question 7	*7 marks*	8
Question 8	*7 marks*	9
Question 9	*7 marks*	10
Question 10	*8 marks*	11
Question 11	*7 marks*	12
Question 12	*6 marks*	13
Question 13	*9 marks*	14
Question 14	*9 marks*	15
Question 15	*8 marks*	16
Question 16	*11 marks*	17

1. The function f is defined by

$$\mathrm{f}(x) = \frac{x}{x^2 + 1}, \qquad 1 \leqslant x \leqslant 3$$

Find the mean value of f over the interval $1 \leqslant x \leqslant 3$.

Give your answer in the form $p \ln q$, where p is a rational number and q is an integer. **[4]**

2. Show that

$$\int_2^{\infty} \frac{x}{(x^2+5)^{\frac{3}{2}}}\,\mathrm{d}x = \frac{1}{3}$$

You must show the limiting process clearly. **[5]**

3. By first expressing

$$\frac{4}{x(x+1)(x+2)}$$

in partial fractions, show that the improper integral

$$\int_1^\infty \frac{4}{x(x+1)(x+2)}\,\mathrm{d}x$$

converges, and find its exact value in the form $\ln k$, where k is a rational number to be determined. **[8]**

4. Let

$$y = \frac{1}{1 - \sin x}$$

By first writing y in terms of $\sec x$ and $\tan x$, find the exact mean value of y with respect to x over the interval $0 \leqslant x \leqslant \frac{1}{6}\pi$. **[6]**

5. The power drawn, p watts, by an electric motor at time t hours after it is switched on can be modelled by the equation

$$p = 40\tanh^2\left(\tfrac{1}{2}t\right), \qquad t \geqslant 0$$

Using an appropriate hyperbolic identity, find, according to the model, the **mean** power drawn by the motor over the first 2 hours after it is switched on. Give your answer as an exact value in the form $\frac{a}{e^2+b}$, where a and b are integers. **[6]**

6. **(a)** Explain why

$$\int_e^\infty \frac{\ln x}{x^3}\,\mathrm{d}x$$

is an improper integral. **[1]**

(b) Use integration by parts to find

$$\int \frac{\ln x}{x^3}\,\mathrm{d}x$$ **[3]**

(c) Hence show that

$$\int_e^\infty \frac{\ln x}{x^3}\,\mathrm{d}x$$

converges, and find its exact value. You should show the limiting process used, and state clearly the behaviour of any limit you rely on. **[4]**

7. **(a)** Express $9x^2 - 18x + 25$ in the form $a(x + p)^2 + q$, where a, p and q are integers. **[2]**

(b) Hence use a standard integral from the formula book to find

$$\int \frac{1}{\sqrt{9x^2 - 18x + 25}}\,\mathrm{d}x$$ **[2]**

(c) Determine the exact mean value of the function

$$\mathrm{g}(x) = \frac{1}{\sqrt{9x^2 - 18x + 25}}, \qquad 0 \leqslant x \leqslant 2$$

giving your answer in the form $A \ln B$, where A and B are rational constants with B an integer. **[3]**

8. Let $y = \ln(1 + x)$.

(a) Show that the mean value of y with respect to x over the interval $0 \leqslant x \leqslant e - 1$ is

$$\frac{1}{e-1}$$ **[4]**

(b) Find, in exact form, the mean value of x with respect to y over the interval $0 \leqslant y \leqslant 1$. **[3]**

9. In an electrical circuit, the current I amps at time t seconds is modelled by

$$I = a\sin nt + b\sin 2nt$$

where a, b and n are positive constants. The RMS value of the current, in amps, is defined to be the square root of the mean value of I^2 over the interval $0 \leqslant t \leqslant \frac{2\pi}{n}$.

Show that the RMS value of the current is

$$\sqrt{\frac{a^2+b^2}{2}}$$

amps. **[7]**

10. Prove that

$$\int_1^{\infty} \frac{x+3}{x^2(x+1)}\,\mathrm{d}x = 3 - \ln 4$$

Show the limiting process clearly. **[8]**

11. **(a)** Explain why

$$\int_0^\infty \frac{10}{(x+2)(3x+1)}\,\mathrm{d}x$$

is an improper integral. **[1]**

(b) Show that

$$\int_0^\infty \frac{10}{(x+2)(3x+1)}\,\mathrm{d}x = a\ln b$$

where a is an integer and b is a rational number, both to be determined. You must show clearly how you deal with the infinite limit. **[6]**

12. **(a)** Explain why

$$\int_0^3 \frac{x}{\sqrt[3]{x^2-1}}\,\mathrm{d}x$$

is an improper integral. **[1]**

(b) Use an appropriate limit argument to show that this integral converges, and find its exact value. **[5]**

13. **(a)** Explain why

$$\int_0^e \frac{(\ln x)^2}{\sqrt{x}}\,\mathrm{d}x$$

is an improper integral. **[1]**

(b) By using integration by parts twice, show that

$$\int x^{-\frac{1}{2}}(\ln x)^2\,\mathrm{d}x = 2\sqrt{x}\left((\ln x)^2 - 4\ln x + 8\right) + c$$ **[4]**

(c) You may assume that, for any constant $k > 0$ and any positive integer n, $x^k(\ln x)^n \to 0$ as $x \to 0^+$.

Show that

$$\int_0^e \frac{(\ln x)^2}{\sqrt{x}}\,\mathrm{d}x$$

exists and find its exact value. **[4]**

14. **(a)** Find

$$\int_1^a \frac{(\ln x)^2}{x^2}\,\mathrm{d}x$$

where $a > 1$, giving your answer in terms of a. **[6]**

(b) Write down the value of

$$\lim_{a\to\infty} \frac{(\ln a)^k}{a}$$

where k is a positive constant. **[1]**

(c) Hence find

$$\int_1^\infty \frac{(\ln x)^2}{x^2}\,\mathrm{d}x$$ **[2]**

15. Show that

$$\int_0^\infty \frac{5x-3}{(x^2+3)(x+1)}\,\mathrm{d}x = p\sqrt{3}\,\pi + \ln q$$

where p and q are rational numbers to be found. **[8]**

16. **(a)** Explain why

$$\int_0^1 (\ln y)^2 \, \mathrm{d}y$$

is an improper integral. **[1]**

(b) (i) Show that the substitution $y = e^{-x}$ transforms

$$\int_0^1 (\ln y)^2 \, \mathrm{d}y \quad \text{into} \quad \int_0^\infty x^2 e^{-x} \, \mathrm{d}x$$

explaining clearly how the limits of integration are obtained. **[3]**

(ii) Evaluate

$$\int_0^\infty x^2 e^{-x} \, \mathrm{d}x$$

showing the limiting process used. **[6]**

(iii) Hence write down the value of

$$\int_0^1 (\ln y)^2 \, \mathrm{d}y$$ **[1]**